



University of Science and Technology · KIMS Campus (UST-KIMS)

Seeking **Future Researchers** to Design Materials with AI

AIMAP Lab – AI for Materials & Processing · UST 소재·공정 인공지능 연구실

AIMAP Lab is recruiting future researchers to lead innovation in materials and manufacturing. We build AI that designs metal alloys, reads microstructures, predicts processes, and runs autonomous experiments.

No programming skills? No materials background? No problem!

We welcome ambitious students who are ready to learn and grow with us.

Research Areas



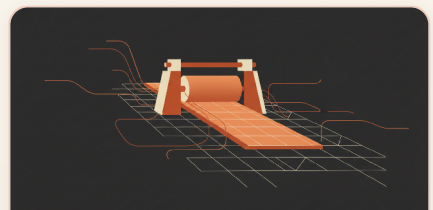
01 AI for Materials Design

Deep-learning property prediction and generative models that accelerate the design and discovery of new alloys.



02 AI-Based Materials Analysis

AI-driven microstructure characterization, automated data analysis, and ML-based performance evaluation.



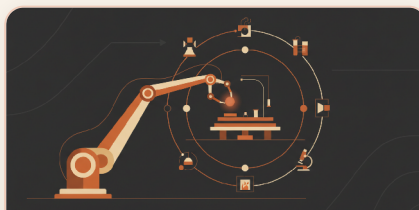
03 Physics-Informed AI for Processes

Hybrid AI-simulation frameworks that embed physical constraints for accurate, reliable process prediction.



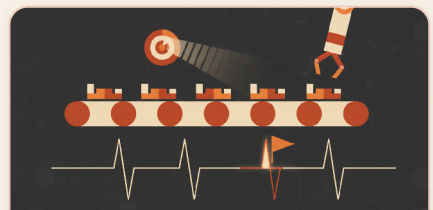
04 Materials VLMs & AI Agents

Vision-language models and AI agents specialized for materials science that automate research workflows.



05 Autonomous Laboratory

Robotics + AI experimental platforms for closed-loop discovery and optimization of metallic materials.



06 Autonomous Manufacturing

Vision- and signal-based anomaly detection and physical AI for manufacturing systems that sense, decide, and act.

77

Journal Articles

61

Conference Talks

38

Funded Projects

3

Open-Source Tools

Principal Investigator



Ho Won Lee, Ph.D. 이호원

Professor, University of Science and Technology (UST)

Director, Division of Materials AI Research, Korea Institute of Materials Science (KIMS)

Ph.D. Mechanical Engineering, KAIST · formerly Max-Planck-Institut für Eisenforschung (MPIE, Germany)

Editorial Board, *Discover Metals* (Springer Nature) · Editorial Board, KSTP

Members

Postdoc Minh Tien Tran · **Ph.D.** Hoang Cuong Phan · Mooyeong Joo · Hoang Hai Nam Nguyen

M.S. Phan Nguyen Duc Hieu · **Intern** Byeong Se Choi

Graduate Student Benefits

- ✓ **Full Support for Conference Travel**
Registration and travel fully covered – recent talks at ICML and ICCV workshops
- ✓ **Flexible, Autonomous Research Environment**
Free use of annual leave, four major insurances included
- ✓ **Mentoring for Top-Tier Journals**
23 journal papers in the last two years (incl. IF 14.0)
- ✓ **Research at a Government Institute (KIMS)**
Hands-on access to advanced equipment and a large GPU cluster

Monthly (gross)	1st year	2nd year+
M.S.	2,125,000 KRW	2,375,000 KRW
Ph.D.	2,635,000 KRW	2,945,000 KRW

UST student researcher rate (as of 2026.07.29)

Selected Publications – PI as Lead/Corresponding Author

- IF 11.4 top 1.9%** Quantifying microstructure- and deformation-mode-dependent heterogeneity using crystal plasticity simulations
International Journal of Mechanical Sciences (2026) · doi.org/10.1016/j.ijmecsci.2026.111818
- IF 11.5 top 2.0%** Parameter-aware high-fidelity microstructure generation using stable diffusion
Advanced Engineering Informatics (2026) · doi.org/10.1016/j.aei.2025.104080
- IF 9.0 top 3.1%** Synthetic microstructure generation using diffusion transformers with structure-preserving augmentation
Engineering Applications of Artificial Intelligence (2026) · doi.org/10.1016/j.engappai.2026.114543
- IF 6.2 top 6.8%** Microstructure-informed forming limits in aluminum alloys: Experiments and crystal plasticity simulations
Materials Characterization (2025) · doi.org/10.1016/j.matchar.2025.115831
- IF 8.2 Q1** Multi-fidelity physics-guided graph neural networks for structure-property relationships in polycrystalline metals
Materials & Design (2026) · doi.org/10.1016/j.matdes.2026.116310

Flagship Projects

- PI** Foundation-model AI service for automated design of metal materials & processes
NST (MSIT) · 2026–2030 · KRW 30.5B total
- PI** Digital innovation platform for metal materials manufacturing
KIAT (MOTIE) · 2022–2026 · KRW 23.2B total
- Co-PI** Aerospace 3D-printing materials database & virtual engineering platform
KIAT (MOTIE) · 2025–2029 · KRW 1.2B total

Research Infrastructure

 GPU Cluster (18× A100 · 24× A6000)	 Automatic Rolling Mill	 Auto Tensile Machine	 X-Ray Diffractometer	 Collaborative Robot	 Large-Format SLA Printer
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We produce process and property data in-house on the rolling mill, tensile machine, and XRD, automate experiments with the robot arm and SLA printer, and train AI models on our GPU cluster – from data generation to validation in one lab.

Apply Now – Grow With Us

Recruiting year-round: **M.S./Ph.D. students (UST)** incl. the integrated program, **postdoctoral researchers**, and **interns**.

Your willingness to learn matters more than your background. Feel free to reach out by email.

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